NANYANG TECHNOLOGICAL UNIVERSITY

SEMESTER I EXAMINATION 2020-2021

MH2500 – Probability and Introduction to Statistics

December 2020 TIME ALLOWED: 2 HOURS

INSTRUCTIONS TO CANDIDATES

1. This examination paper contains **SIX (6)** questions and comprises **FOUR (4)** printed pages including appendix.
2. Answer **ALL** questions. The marks for each question are indicated at the beginning of each question.
3. Answer each question beginning on a **FRESH** page of the answer book.
4. This is a **RESTRICTED OPEN BOOK** exam. Candidates are allowed **ONE** double-sided A4 size help sheet.
5. Candidates may use calculators. However, they should write down systematically the steps in the workings.
6. The table with values of the standard normal distribution is included at the end of this examination paper.

QUESTION 1 **(20 marks)**

In a certain assembly plant, three machines, B_1, B_2 and B_3 make 25%, 35%, and 40%, respectively, of the products. It is known from the past experience that 5%, 4%, and 2% of the products made by each machine, respectively, are defective. Now, suppose a finished product is randomly selected.

(a) Find the probability that the selected product is a defective.

(b) Assume the selected product is a defective. Find the probability that this product was made by Machine B_1.

(c) If we know that the selected product is not produced by B_3. Find the probability that it is produced by B_1.

Solution: Let D, B_1, B_2 and B_3 be the following events.

- D: the product is defective,
- B_i: the product is made by Machine B_i, $i = 1, 2, 3$.

From the information provided, we have

$$P(B_1) = 0.25, \quad P(B_2) = 0.35, \quad P(B_3) = 0.4,$$

and hence

$$P(D|B_1) = 0.05, \quad P(D|B_2) = 0.04, \quad P(D|B_3) = 0.02.$$

(a) $P(D) = P(B_1)P(D|B_1) + P(B_2)P(D|B_2) + P(B_3)P(D|B_3)$

$$= (0.25)(0.05) + (0.35)(0.04) + (0.40)(0.02) = 0.0345.$$

(b) Our aim is to find $P(B_1|D)$, and we apply Bayes' Rule here.

$$P(B_1|D) = \frac{P(B_1)P(D|B_1)}{P(D)} = \frac{(0.25)(0.05)}{0.0345} = \frac{125}{345} = \frac{25}{69} = 0.3623.$$

(c) $P(B_1|B_3') = \dfrac{P(B_1 \cap B_3')}{P(B_3')} = \dfrac{P(B_1)}{P(B_1 \cup B_2)} = \dfrac{0.25}{0.25 + 0.35} = \dfrac{5}{12} = 0.4167.$

QUESTION 2. **(15 Marks)**

The serum cholesterol level X in 14-year-old boys has approximately a normal distribution with mean 170 and standard deviation 30.

(a) Find the probability that the serum cholesterol level of a randomly chosen 14-year-old exceeds 230.

(b) In a middle school there are 300 14-year-old boys. Find the probability that at least 8 boys have a serum cholesterol level that exceeds 230.

Solution: Note that $Z = \dfrac{X - 170}{30}$ is a standard normal variable.

(a) The required probability is $P(X > 230)$ which is evaluated as follows:

$$P(X > 230) = P\big(Z > \frac{230 - 170}{30}\big) = P(Z > 2)$$
$$= 1 - P(Z < 2) = 1 - 0.9772 = 0.0228.$$

(b) In a middle school there are 300 14-year-old boys.

Let Y be the number of boys out of 300 who have a serum cholesterol level that exceeds 230. Then Y is a binomial random variable with $n = 300$ and $p = 0.0228$ (from part (a).) Its mean $\mu_Y = np = 300(0.0228) = 6.84$ and its variance

$$\sigma_Y^2 = np(1-p) = 300(0.0228)(1 - 0.0228) = 6.684.$$

Thus, $\sigma_Y = 2.585$.

We use normal approximation for the distribution of Y, with continuity correction. Thus $Z = \dfrac{Y - 6.84}{2.585}$ approximates a standard normal variable.

The probability that at least 8 boys have a serum cholesterol level that exceeds 230 is given by

$$P(Y \geq 8) \approx P\big(Z \geq \frac{7.5 - 6.84}{2.585}\big) = P(Z \geq 0.26) = P(Z < -0.26) = 0.3974.$$

QUESTION 3. **(15 Marks)**

The joint density function of X and Y is given as:

$$f(x, y) = \begin{cases} \dfrac{1}{y} & \text{if } 0 < x < y, 0 < y < 1, \\ 0 & \text{elsewhere.} \end{cases}$$

(a) Find the marginal distributions of X and Y, respectively.

Solution: $g(x) = \int_x^1 \frac{1}{y} dy = [\ln y]\Big|_x^1 = \ln 1 - \ln x = -\ln x.$

$h(y) = \int_0^y \frac{1}{y} dx = \frac{1}{y} \int_0^y 1 dx = 1.$

(b) Are these two random variables X and Y independent?

Solution: No. As for (x, y) in the region, $f(x, y) = 1/y$ and $g(x)h(y) = -\ln x \cdot 1 = -\ln x$, and $f(x, y) \neq g(x)h(y)$.

(c) Find the probability $P\left\{X + Y > \frac{1}{2}\right\}$.

Solution: Find the region with $X + Y > 1/2$. Then

$$P\left\{X + Y > \frac{1}{2}\right\} = 1 - \frac{1}{2}\ln 2.$$

QUESTION 4 **(20 marks)**

The fraction X of male runners and the fraction Y of female runners who compete in marathon races are described by the joint density function

$$f(x, y) = \begin{cases} 8xy & 0 \leq y \leq x \leq 1, \\ 0 & \text{elsewhere.} \end{cases}$$

1. Find the covariance of X and Y.
2. Find the correlation coefficient of X and Y.

Solution: We first compute the marginal density functions. They are

$$g(x) = \begin{cases} 4x^3 & 0 \leq x \leq 1, \\ 0 & \text{elsewhere,} \end{cases}$$

and

$$h(y) = \begin{cases} 4y - 4y^3 & 0 \le y \le 1, \\ 0 & \text{elsewhere}, \end{cases}$$

From these marginal density function, we compute

$$E(X) = \int_0^1 \left(x \cdot 4x^3\right) dx = \frac{4}{5},$$

and

$$E(Y) = \int_0^1 \left(y(4y - 4y^3)\right) dy = 4 \times \frac{1}{3} - 4 \times \frac{1}{5} = \frac{8}{15}.$$

From the joint density function given, we have

$$E(XY) = \int_0^1 \int_y^1 \left((xy)(8xy)\right) dx dy = \int_0^1 \left(8y^2 \int_y^1 x^2 dx\right) dy$$

$$= \int_0^1 \left(8y^2 \times \frac{1}{3}(1 - y^3)\right) dy = \frac{8}{3} \int_0^1 \left(y^2 - y^5\right) dy = \frac{8}{3}\left(\frac{1}{3} - \frac{1}{6}\right) = \frac{4}{9}.$$

Thus,

$$\sigma_{XY} = E(XY) - E(X)E(Y) = \frac{4}{9} - \frac{4}{5} \times \frac{8}{15} = \frac{4}{225}.$$

QUESTION 5 **(15 marks)**

An urn has n white and m black balls that are removed one at a time in a randomly chosen order. Find the expected number of instances in which a white ball is immediately followed by a black one.

[Solution:] Let I_j equal 1 if the j-th ball withdrawn is white and the $(j+1)$-st is black, and let I_j equal 0 otherwise. If X is the number of instances in which a white ball is immediately followed by a black one, then we may express X as

$$X = \sum_{j=1}^{n+m-1} I_j.$$

Thus,

$$\mathbb{E}(X) = \sum_{j=1}^{n+m-1} \mathbb{E}(I_j) = \sum_{j=1}^{n+m-1} P\{j\text{-th selection is white, } (j+1)\text{-st is black}\}$$

$$= \sum_{j=1}^{n+m-1} P\{j\text{-th selection is white}\}P\{(j+1)\text{-st is black}|j\text{-th selection is white}\}$$

$$= \sum_{j=1}^{n+m-1} \frac{n}{n+m}\frac{m}{n+m-1} = \frac{nm}{n+m}.$$

QUESTION 6 **(15 marks)**

An instructor has 50 exams that will be graded in sequence. The times required to grade the 50 exams are independent, with a common distribution that has mean 20 minutes and standard deviation 4 minutes. By applying the Central Limit Theorem, approximate the probability that the instructor will grade at least 25 of the exams in the first 450 minutes of work.

Solution: Let X_i be the time that it takes to grade exam i, then

$$X = \sum_{i=1}^{25} \mathbb{E}(X_i)$$

is the time it takes to grade the first 25 exams. Because the instructor will grade at least 25 exams in the first 450 minutes of work if the time it takes to grade the first 25 exams is less than or equal to 450, we see that the desired probability is $P\{X \leq 450\}$. To approximate this probability, we use the central limit theorem. Now,

$$\mathbb{E}(X) = \sum_{i=1}^{25} \mathbb{E}(X_i) = 25(20) = 500$$

and

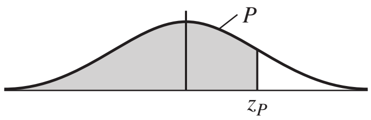
$$Var(X) = \sum_{i=1}^{25} Var(X_i) = 25(16) = 400.$$

Consequently, with Z being a standard normal random variable, we have

$$P\{X \leq 450\} = P\left\{\frac{X - 500}{\sqrt{400}} \leq \frac{450 - 500}{\sqrt{400}}\right\}$$

$$\approx P\{Z \leq -2.5\} = P\{Z \geq 2.5\} = 1 - \Phi(2.5) \approx 0.06.$$

TABLE Cumulative Normal Distribution—Values of *P* Corresponding to z_p for the Normal Curve



z is the standard normal variable.

z_p	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

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